

CRISP-DM: Business Understanding

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Spring, 2026

This report states my business understanding side of the CRISP-DM methodology reference model with regard to the provided [dataset](#).

1 Determine Business Objectives

1.1 Background

In this setting, I propose to study a publisher company that is interested in investigation of the Steam market to discover promising developers for production contracts.

1.2 Business Objectives

The business aims to find emerging developer teams that recently started their careers and have few or no contracts with major game publishers. While these developers lack a brand, some teams have the potential to develop successful products. Since investing in an unproven team can be risky, a data-driven approach is necessary to address clearly defined business needs. Therefore, the following key business goals must be achieved:

1. *Identify the characteristics of developers (based on their application portfolios) that correlate with having at least one published game with publishers in the top 100 by published volume on Steam.*
2. *Create a scoring system that ranks emerging developers (first release within 3 years) according to similarity to the successful partnership patterns identified.*

1.3 Business Success Criteria

The following business metric determines the success for this case:

1. The model reveals at least three distinct and interpretable developer attributes (e.g., portfolio size, pricing, player engagement) that are strongly associated with top publisher partnerships, as evidenced by consistent high importance across models.

2. The system detects at least 5 emerging developers (first release within 3 years, no previous contracts with the top 100 publishers by published volume on Steam) that have enough evidence ($> 70\%$) to secure a partnership with a top publisher.

2 Assess Situation

2.1 Inventory of Resources

1. **Personnel:** data analyst;
2. **Data:** Steam Dataset 2025 on Kaggle;
3. **Computing Resources:** Innopolis data hub servers;
4. **Software:** Python interpreter with data science and ML libraries.

2.2 Requirements, Assumptions, and Constraints

- R:** Given the availability of the dataset, the project is expected to be finished in a month to start negotiations before the end of the first quarter of 2026. The product of the work should be a verbal suggestion based on correlation represented as a composition of properties of a developer team. For example, the report can indicate that *"a team of developers with mean average game score on Steam not less than 85% and at least 5 published games secures a partnership with a major game publisher with 80% probability"* Furthermore, the list of at least 5 developer teams suitable for the company needs is to be reported. The analysis is to be conducted according to Steam data collection policies to ensure the legality and security of public appearance.
- A:** The major assumption behind this project is the reliable and sufficient information about partnerships between developers and publishers on Steam. Additionally, the dataset is considered suitable for ranking publishing companies. Generally, the provided data are considered relevant and useful for the needs of the business.
- C:** The project is confined with the resources of the data hub servers. The search of additional information can be performed only through open access gateways of the web.

The analysis identifies correlations between developer characteristics and partnership status but cannot establish causality. Results indicate which characteristics are associated with successful partnerships, not that these characteristics cause partnerships. Business decisions should consider these patterns as a recommendation for further inspection.

2.3 Risks and Contingencies

The following table ranks all risks (starts from highest priority) and lists possible actions upon them:

Rank	Risk	Contingencies
1	Selected dataset is not relevant	Start searching to find a new dataset within a week; if not found, find agency to purchase relevant data
2	Part of dataset is not relevant and/or missing	Prioritize missing information and start additional data collection from the web.
3	Modeling fails to achieve business goals	Revisit business understanding
4	Issues with accessing servers of data hub	Call service providers and use other provider if the recovery takes more than 3 days
5	Personnel is unable to finish their job	Outsource the work to be done

Table 1: Ranked Risks and Contingencies

2.4 Terminology

- **Publisher:** An entity that funds developers to create applications on Steam.
- **Developer:** An entity that creates applications on Steam.
- **Emerging Developer:** A developer whose first Steam application was released within 3 years.
- **Top Publisher Partnership:** Co-occurrence of a developer and a publisher, where the publisher is in the top 100 by published volume on Steam.
- **Top 100 Publishers:** the first 100 publishers having at least 10 partnerships with developers sorted by published volume on Steam in descending order.
- **Partnership Prediction:** The model’s estimation that a developer without current top publisher partnerships would secure one.
- **Partnership-density Clusters:** Groups of developers sharing common features with assigned estimated proportion of teams who secured a top publisher partnership (partnership density).
- **Precision:** The proportion of developers predicted to secure partnerships who historically did so within the specified time frame.

2.5 Costs and Benefits

The costs are estimated based on the project’s duration (one month) and the resources listed in the inventory:

- **Personnel:** one data analyst, full time for one month. Estimated cost: \$5,000 (assuming a monthly rate of \$5,000).
- **Computing:** access to university data hub servers (no additional cost) plus a contingency for a cloud instance if needed: \$200.
- **Data:** the Kaggle dataset is free; if additional data were required, an external purchase could cost up to \$1,000 (not used in this project).
- **Contingency:** \$1,000 (10% of personnel cost) for unforeseen issues.

Total estimated cost: \$6,200 (upper bound \$7,200 if external data needed).

The benefits are assessed through scenario analysis based on the success criteria:

- **Low scenario:** The system identifies 5 emerging developers, but none sign a contract. Benefit: \$0 immediate, but the company gains a reusable analytical framework and market insights.
- **Medium scenario:** One of the recommended developers signs a contract. A typical development contract with an emerging team might generate \$50,000–\$100,000 in net revenue over two years. Benefit: \$50,000.
- **High scenario:** Two or more recommended developers sign contracts, or a single contract leads to a highly successful game. Benefit: \$150,000+.

The project’s expected value is positive even in the medium scenario, and the cost is modest compared to the potential upside. The analysis is bounded by the assumption that the company would otherwise spend similar resources on manual scouting with lower efficiency.

3 Determine Data Mining Goals

3.1 Data Mining Goals

Translation of the business goals in technical terms will result in the following objectives:

1. *Design features for describing developer companies from application history and product reviews, including portfolio characteristics (size, genre diversity, etc.), game quality metrics (average review score, review volume), and temporal patterns (years since first release).*
2. *Perform clustering analysis to identify natural groupings of developers based on identified features.*

3. *Train a classification model to predict partnership status, using the clusters as additional features and analyzing feature importance to identify key success factors.*
4. *Rank emerging developers according to the classification probability, filtering for those without current top-publisher partnerships.*

3.2 Data Mining Success Criteria

1. At least 10 features covering application history and product reviews are identified.
2. Clustering yields interpretable segments (silhouette score > 0.25) with at least one cluster showing $> 10\%$ partnership density.
3. The classification model achieves ROC-AUC > 0.8 with at least 3 features whose importance (measured via built-in feature importance or SHAP) is at least 1% of the maximum importance among all features.
4. The model is well calibrated, as measured by a Brier score < 0.1 and a calibration curve whose maximum deviation from the diagonal (expected calibration error, ECE) is < 0.1 .
5. The classification model achieves precision > 0.6 on the test set (at the threshold tuned on validation data). The recall on the test set is expected to be in the range 0.25 ± 0.05 .

4 Produce Project Plan

4.1 Project Plan

I propose the following plan for this project:

Task	Week	Input	Output	Resources	Evaluation Criteria
Data Understanding	Week 1	Dataset	EDA Report with proposed features	Python Interpreter	(1) Correspondence analysis with business understanding; (2) Feature engineering success criterion (№1)
Data Preparation	Week 2	Dataset, features	Task-oriented dataset	Python Interpreter	100% data availability
Clusterization Modelling	Week 3	Task-oriented dataset	(1) Clusterization report including clusters identified and partnership density; (2) Augmented task-oriented dataset	Data hub server	Clusterization success criterion (№2)
Classification Modelling	Week 3	Augmented task-oriented dataset	(1) Feature significance; (2) Trained model	Data hub server	Classification success criterion (№3)
Result Showcase	Week 4	Clusterization report and classification model	List of 5 recommended developers	Data hub server	Recommendation list success criterion (№4)

Table 2: Project Plan

4.2 Initial Tools and Techniques Assessment

1. Data Processing: Python (pandas, numpy, matplotlib).
2. Clustering: K-means, DBSCAN with silhouette analysis for validation.
3. Classification: Logistic Regression (interpretability) and Random Forest (performance).
4. Feature Importance: SHAP values for model interpretation.